Movies

marchese.65

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Overview

For this project, my goal is to discover the variables that make a movie great; based on profit, genre, production company, ROI, rating, and more! As well as that, I want to discover historically what has made movies great and make a prediction based on the results I find, what would the rating and ROI be!

This dataset is a large-scale unified movies and TV shows corpus built from real TMDB data, covering 1,478,995 movies and 168,633 TV titles across all content types. Language codes are expanded to full names (ISO 639-1), country codes resolved to full names (ISO 3166-1 a-2), and genres normalised to a shared 23-category taxonomy. Financial data is available for 5.68% of titles; gaps are filled with genre-aware synthesised values and flagged explicitly. The dataset is designed for revenue and rating prediction, hit classification, recommendation systems, and multi-task entertainment market analysis.

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.1
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```
tmbd_movies_cleaned <- tmdb_data_full |>
  filter(data_completeness_score == 1,
         content_type == "movie",
         vote_count > 100,
         budget_usd > 50000,
         primary_genre != "TV Movie",
         primary_genre != "Music") |>
  select(!c(
    popularity, status, adult, tagline, keywords,
    origin_country_expanded, genres_unified, potential_outlier_revenue,
    vote_momentum, popularity_tier, genre_music,
    tv_format:genres_raw,
    is_financial_data_available:is_synthetic_revenue,
    release_month:release_day,
    franchise_value_score:target_hit,
    genre_tv_movie:type_short
  )) |>
  rename(rating_average = vote_average)
```

In order to begin this project, and to remove all outliers, I had to fix the data a bit. In the data set it had a "data_completeness_score". This means that there was information in all the key fields of data for the movie. If a movie didn't have all of them, it was removed from the data set I conducted my research on. I also removed all movies from the data set that had less than 100 votes on the ratings. This removes movies that scored high on the 1-10 scale on the movie rating on TMBD that had such low reviews that they would be outliers. I also filtered the budgets to be greater than 50000 so there were no outlier ROIs.

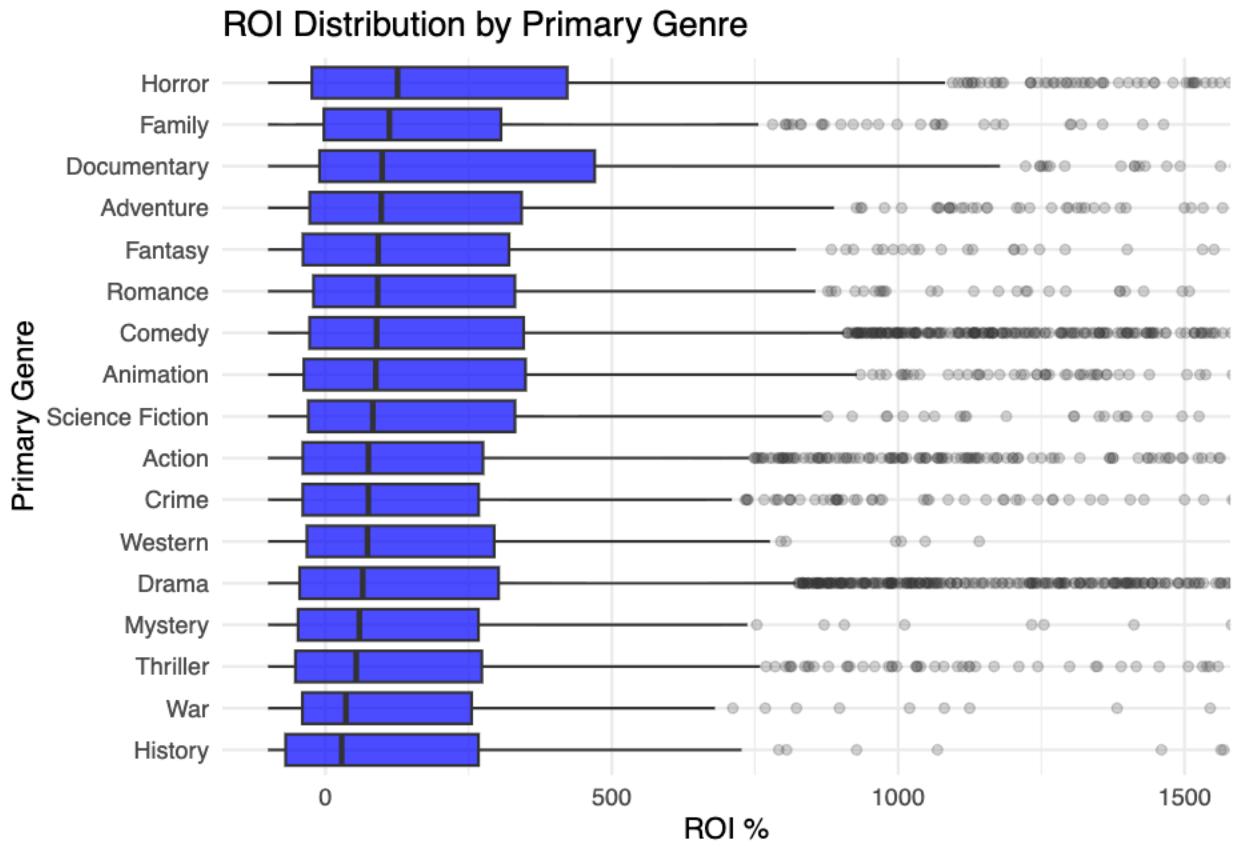
For this project, I am determining top movies statistics by the rating and ROI of them, as these are the most important things in determining a good movie. An important thing to note as we explore different questions, is that the groups with large amounts of data points in them tend to land around an average rating of 6 and 100% ROI, we will attempt to discover changes regardless of this fact.

Questions

Does genre have an effect on ratings and ROI (based on rating and ROI)?

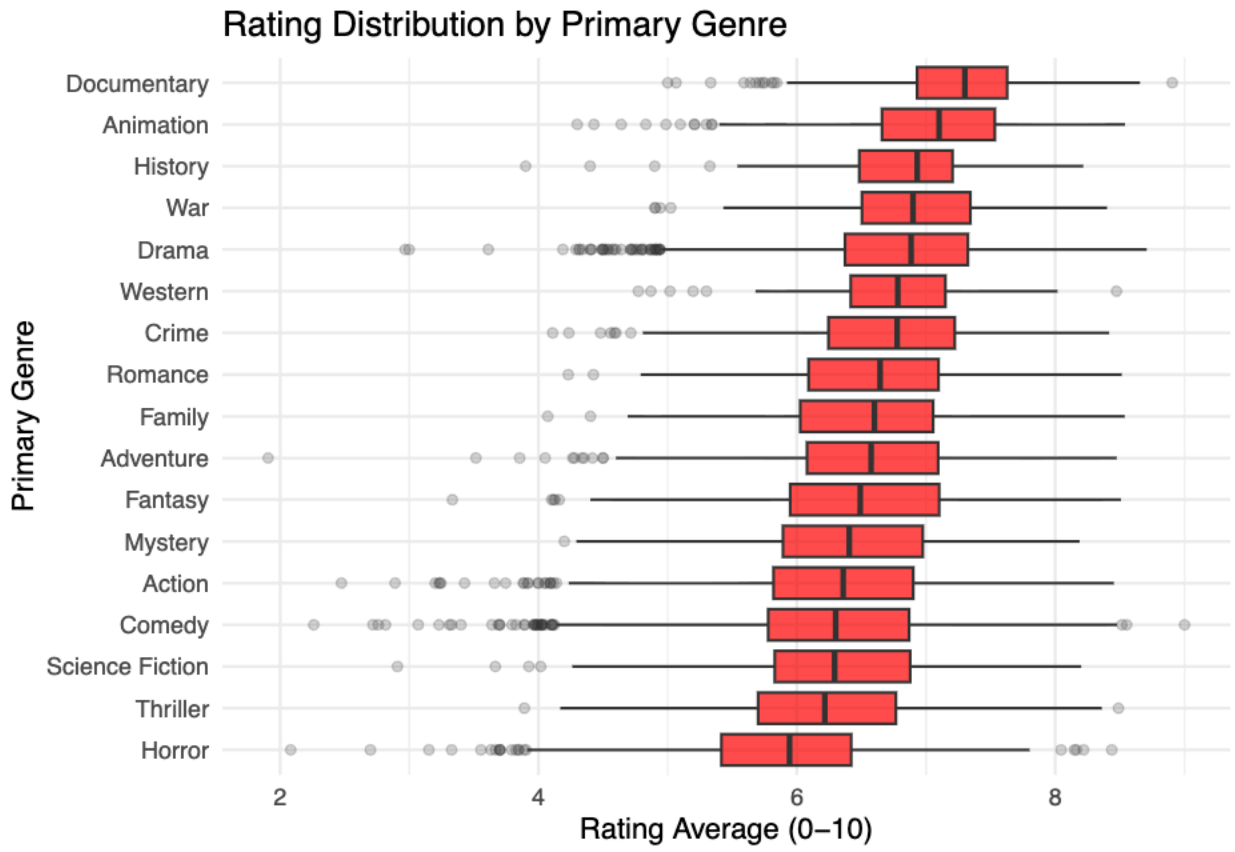
```
roi_plot <- tmbd_movies_cleaned |>
  ggplot(aes(x = reorder(primary_genre, roi_pct, FUN = median), y = roi_pct)) +
    geom_boxplot(outlier.alpha = 0.2, fill = "blue", alpha = 0.7) +
    coord_flip(ylim = c(-100, 1500)) +
    labs(
      title = "ROI Distribution by Primary Genre",
      x = "Primary Genre",
      y = "ROI %"
    ) +
    theme_minimal()

print(roi_plot)
```




```r
rating_plot <- tmbd_movies_cleaned |>
   ggplot(aes(x = reorder(primary_genre, rating_average, FUN = median), y = rating_average)) +
    geom_boxplot(outlier.alpha = 0.2, fill = "red", alpha = 0.7) +
    coord_flip() +
    labs(
      title = "Rating Distribution by Primary Genre",
      x = "Primary Genre",
      y = "Rating Average (0-10)"
    ) +
    theme_minimal()

print(rating_plot)
```



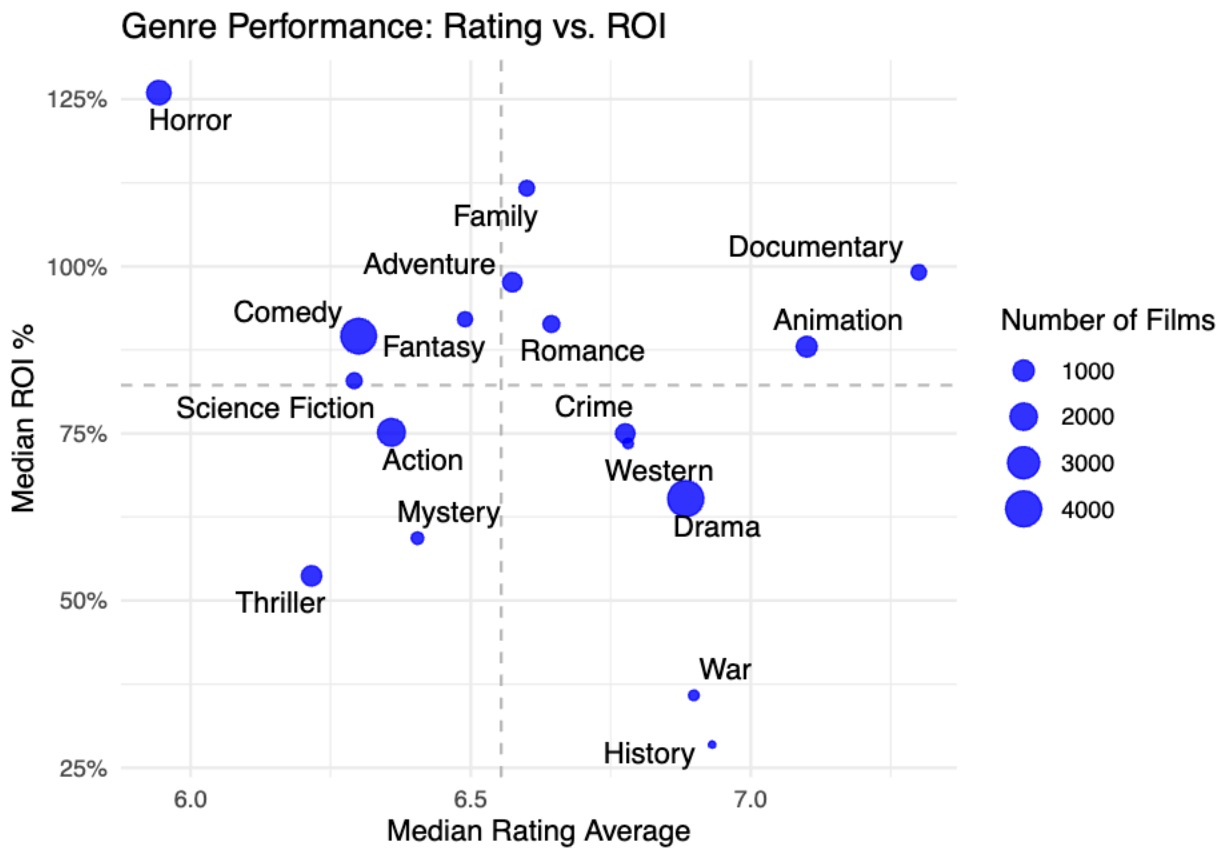

```
library(ggrepel)

genre_summary <- tmbd_movies_cleaned |>
  group_by(primary_genre) |>
  summarise(
    median_rating = median(rating_average, na.rm = TRUE),
    median_roi = median(roi_pct, na.rm = TRUE),
    movie_count = n()
  )


overall_median_rating <- median(tmbd_movies_cleaned$rating_average, na.rm = TRUE)
overall_median_roi <- median(tmbd_movies_cleaned$roi_pct, na.rm = TRUE)

ggplot(genre_summary, aes(x = median_rating, y = median_roi, label = primary_genre)) +
  geom_vline(xintercept = overall_median_rating, linetype = "dashed", color = "gray") +
  geom_hline(yintercept = overall_median_roi, linetype = "dashed", color = "gray") +
  geom_point(aes(size = movie_count), color = "blue", alpha = 0.8) +
  geom_text_repel(size = 3.8, box.padding = 0.35) +
  scale_y_continuous(labels = scales::label_percent(scale = 1)) +
  labs(
    title = "Genre Performance: Rating vs. ROI",
    x = "Median Rating Average",
    y = "Median ROI %",
    size = "Number of Films"
  ) +
```

```
  theme_minimal()
```




```
genre_combined_performance <- tmbd_movies_cleaned |>
  group_by(primary_genre) |>
  summarise(
    movie_count = n(),
    median_rating = median(rating_average, na.rm = TRUE),
    median_roi = median(roi_pct, na.rm = TRUE)
  ) |>
  mutate(
    rating_percentile = percent_rank(median_rating),
    roi_percentile = percent_rank(median_roi),
    combined_genre_score = (rating_percentile + roi_percentile) / 2
  ) |>
  arrange(desc(combined_genre_score))

print(genre_combined_performance, n = Inf)
```

```
## # A tibble: 17 x 7
##    primary_genre   movie_count median_rating median_roi rating_percentile
##    <chr>                 <int>         <dbl>      <dbl>             <dbl>
##  1 Documentary             406           7.3        99.1             1
##  2 Animation               946           7.1        88.0             0.938
##  3 Family                  421           6.6       112.              0.5
##  4 Adventure               786           6.57       97.6             0.438
```

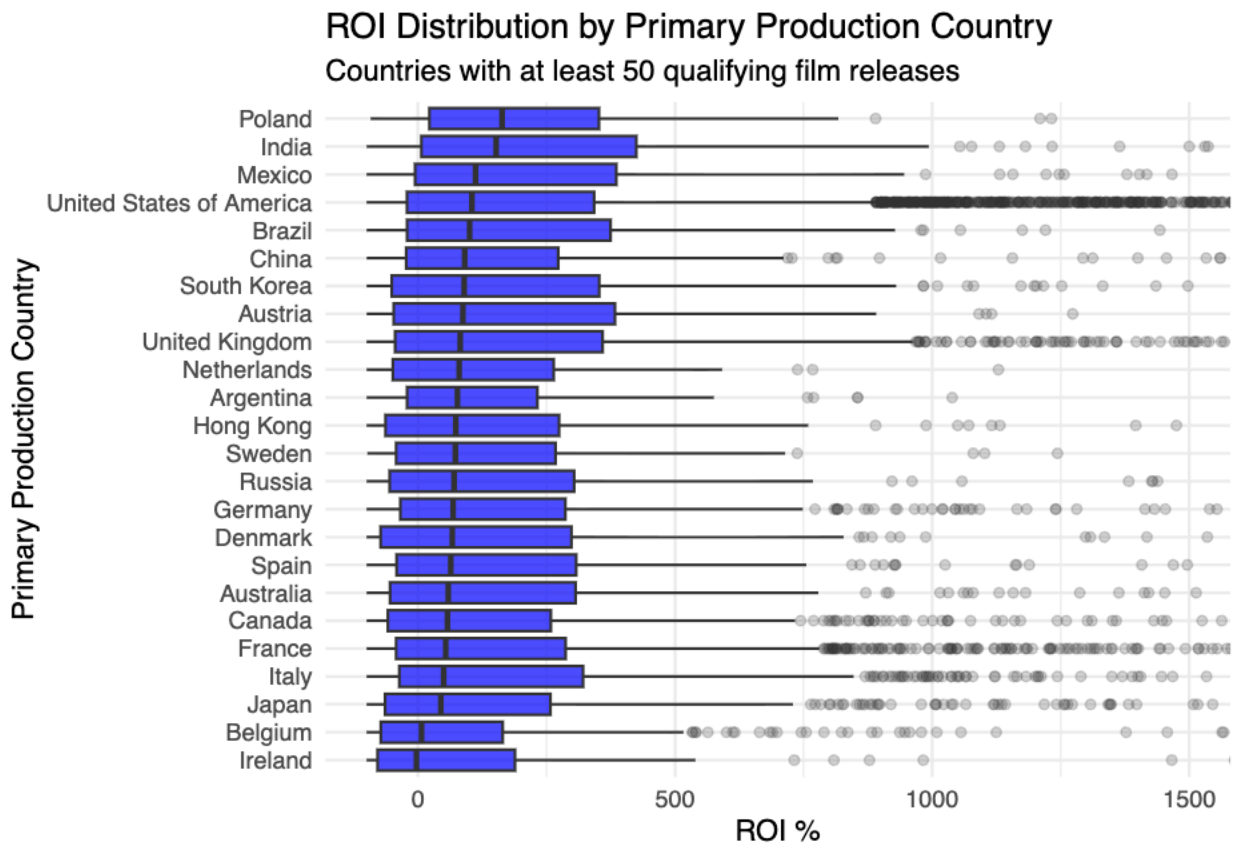
```
##  5 Romance                530          6.64       91.4           0.562
##  6 Fantasy                371          6.49       92.1           0.375
##  7 Crime                  771          6.78       75             0.625
##  8 Drama                 4011          6.88       65.2           0.75
##  9 Horror                1478          5.94      126.            0
## 10 Western                145          6.78       73.5           0.688
## 11 History                105          6.93       28.4           0.875
## 12 War                    150          6.90       35.8           0.812
## 13 Comedy                3892          6.3        89.6           0.188
## 14 Action                2084          6.36       75.2           0.25
## 15 Science Fiction        438          6.29       82.9           0.125
## 16 Mystery                219          6.40       59.3           0.312
## 17 Thriller               881          6.22       53.7           0.0625
## # i 2 more variables: roi_percentile <dbl>, combined_genre_score <dbl>
```

After reviewing the boxplots and quadrant graph, there is a strong correlation between the genre and the effect on rating and ROI. The most interesting results is that while the ROI is extremely high for horror, the highest of all movies, it has the lowest median rating. The best overall genre performance was documentary with the highest median rating and ROI. It is also good to note that the most number of films were comedy and drama, showing these are the most popular films made that fit into our categories that we are testing.

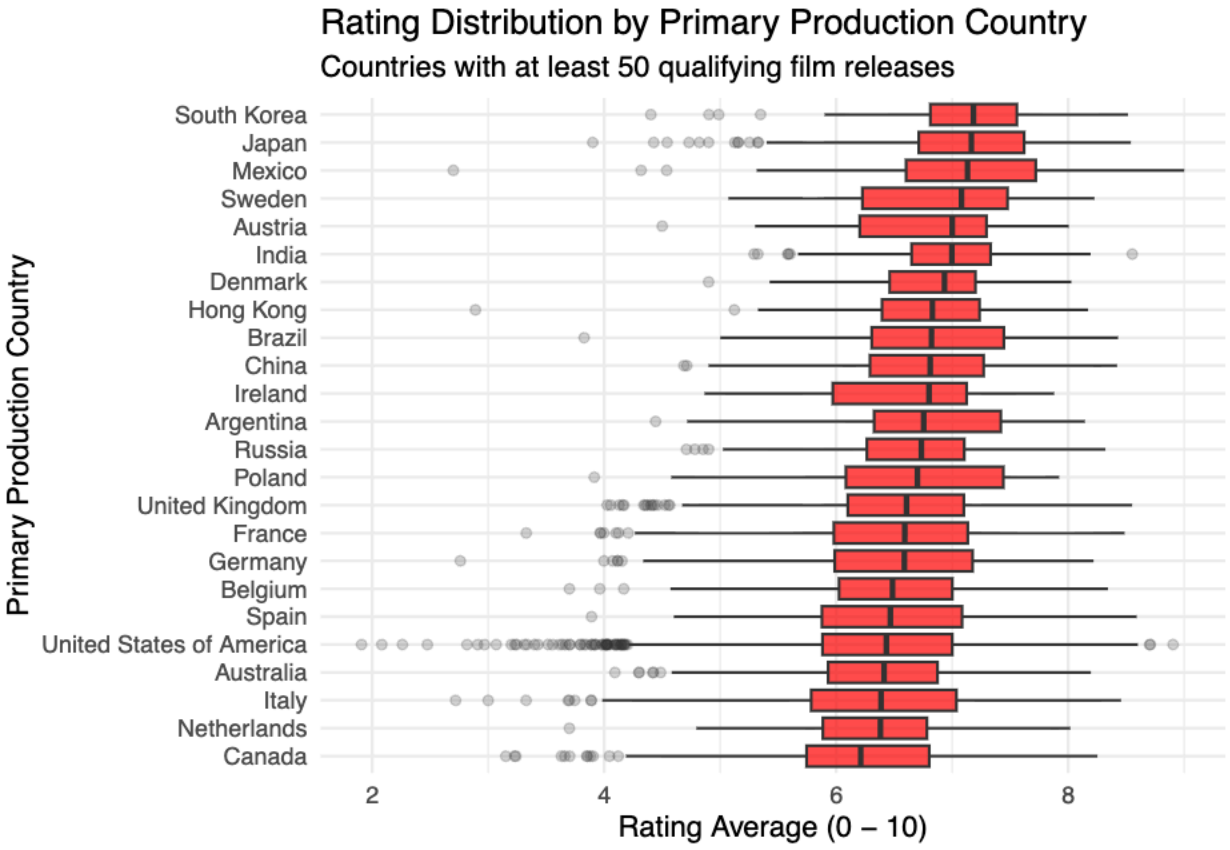
What countries produce the best movies?

```
top_countries_volume <- tmbd_movies_cleaned |>
  filter(!is.na(primary_production_country)) |>
  count(primary_production_country) |>
  filter(n >= 50) |>
  pull(primary_production_country)

tmbd_movies_cleaned |>
  filter(primary_production_country %in% top_countries_volume) |> # Added missing pipe |>
  ggplot(aes(x = reorder(primary_production_country, roi_pct, FUN = median), y = roi_pct)) +
  geom_boxplot(outlier.alpha = 0.2, fill = "blue", alpha = 0.7) +
  coord_flip(ylim = c(-100, 1500)) +
  labs(
    title = "ROI Distribution by Primary Production Country",
    subtitle = "Countries with at least 50 qualifying film releases",
    x = "Primary Production Country",
    y = "ROI %"
  ) +
  theme_minimal()
```




```
tmbd_movies_cleaned |>
  filter(primary_production_country %in% top_countries_volume) |> # Added missing pipe |>
  ggplot(aes(x = reorder(primary_production_country, rating_average, FUN = median), y = rating_average)
  geom_boxplot(outlier.alpha = 0.2, fill = "red", alpha = 0.7) +
  coord_flip() +
  labs(
    title = "Rating Distribution by Primary Production Country",
    subtitle = "Countries with at least 50 qualifying film releases",
    x = "Primary Production Country",
    y = "Rating Average (0 - 10)"
  ) +
  theme_minimal()
```

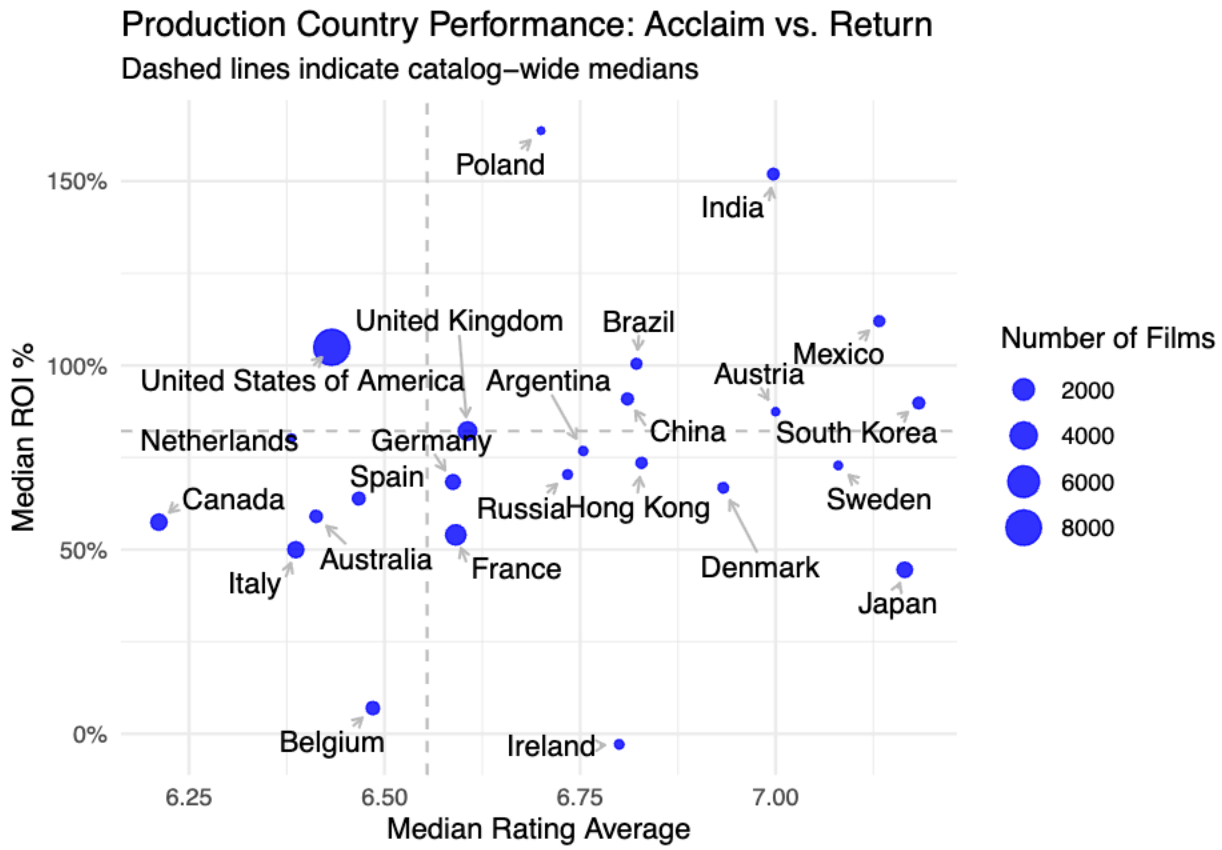



```
primary_production_country_summary <- tmbd_movies_cleaned |>
  filter(!is.na(primary_production_country)) |>
  group_by(primary_production_country) |>
  summarise(
    median_rating = median(rating_average, na.rm = TRUE),
    median_roi = median(roi_pct, na.rm = TRUE),
    movie_count = n()
  ) |>
  filter(movie_count >= 50)

overall_median_rating <- median(tmbd_movies_cleaned$rating_average, na.rm = TRUE)
overall_median_roi <- median(tmbd_movies_cleaned$roi_pct, na.rm = TRUE)

ggplot(primary_production_country_summary, aes(x = median_rating, y = median_roi, label = primary_produc
  geom_vline(xintercept = overall_median_rating, linetype = "dashed", color = "gray") +
  geom_hline(yintercept = overall_median_roi, linetype = "dashed", color = "gray") +
  geom_point(aes(size = movie_count), color = "blue", alpha = 0.8) +
  geom_text_repel(
    size = 3.8,
    box.padding = 0.5,
    point.padding = 0.3,
    min.segment.length = 0,
    segment.color = "grey",
    segment.size = 0.5,
    arrow = arrow(length = unit(0.015, "npc"))
    ) +
```

```
  scale_y_continuous(labels = scales::label_percent(scale = 1)) +
  labs(
    title = "Production Country Performance: Acclaim vs. Return",
    subtitle = "Dashed lines indicate catalog-wide medians",
    x = "Median Rating Average",
    y = "Median ROI %",
    size = "Number of Films"
  ) +
  theme_minimal()
```




```
country_combined_performance <- tmbd_movies_cleaned |>
  group_by(primary_production_country) |>
  summarise(
    movie_count = n(),
    median_rating = median(rating_average, na.rm = TRUE),
    median_roi = median(roi_pct, na.rm = TRUE)
  ) |>
  filter(movie_count >= 50) |>
  mutate(
    rating_percentile = percent_rank(median_rating),
    roi_percentile = percent_rank(median_roi),
    combined_genre_score = (rating_percentile + roi_percentile) / 2
  ) |>
  arrange(desc(combined_genre_score))
```

```r
print(country_combined_performance, n = Inf)
```

```
## # A tibble: 24 x 7
##    primary_production_c~1 movie_count median_rating median_roi rating_percentile
##    <chr>                         <int>         <dbl>      <dbl>             <dbl>
##  1 Mexico                          158          7.13     112.              0.913
##  2 India                           206          7.00     152.              0.783
##  3 South Korea                     219          7.18      89.8             1
##  4 Austria                          53          7         87.5             0.826
##  5 Brazil                          149          6.82     100.              0.652
##  6 Poland                           51          6.7      164.              0.435
##  7 China                           230          6.81      90.9             0.609
##  8 Sweden                           60          7.08      72.8             0.870
##  9 Hong Kong                       176          6.83      73.5             0.696
## 10 Argentina                        77          6.75      76.8             0.522
## 11 Denmark                         136          6.93      66.8             0.739
## 12 Japan                           692          7.16      44.5             0.957
## 13 United Kingdom                 1255          6.61      82.2             0.391
## 14 United States of Amer~         8344          6.43     105.              0.174
## 15 Russia                          88          6.73      70.4             0.478
## 16 Germany                        580          6.59      68.4             0.304
## 17 Netherlands                     56          6.38      80.3             0.0435
## 18 Ireland                         83          6.8       -2.87            0.565
## 19 France                         1695          6.59      54              0.348
## 20 Spain                           323          6.47      63.9             0.217
## 21 Australia                       284          6.41      59.0             0.130
## 22 Belgium                         379          6.48       6.96            0.261
## 23 Canada                          840          6.21      57.5             0
## 24 Italy                           792          6.39      50.0             0.0870
## # i abbreviated name: 1: primary_production_country
## # i 2 more variables: roi_percentile <dbl>, combined_genre_score <dbl>
```

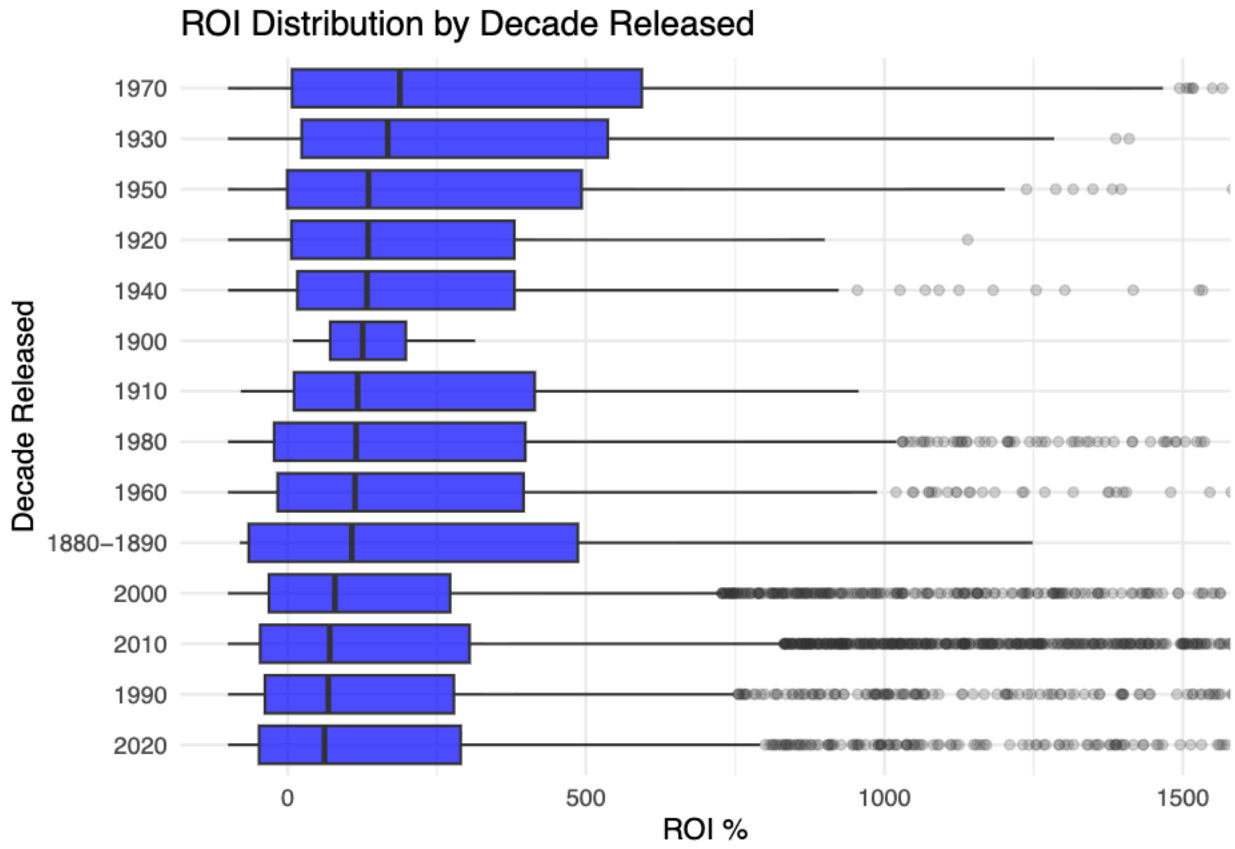
In order to answer the question of which countries produce the best movies I removed all countries with less than 50 movies made. The results of this is so surprising. The top filmed in countries are South Korea, Mexico, and India. The worst countries, by a wide margin are Italy, Canada, and Belgium.

What decade has produced the best movies?

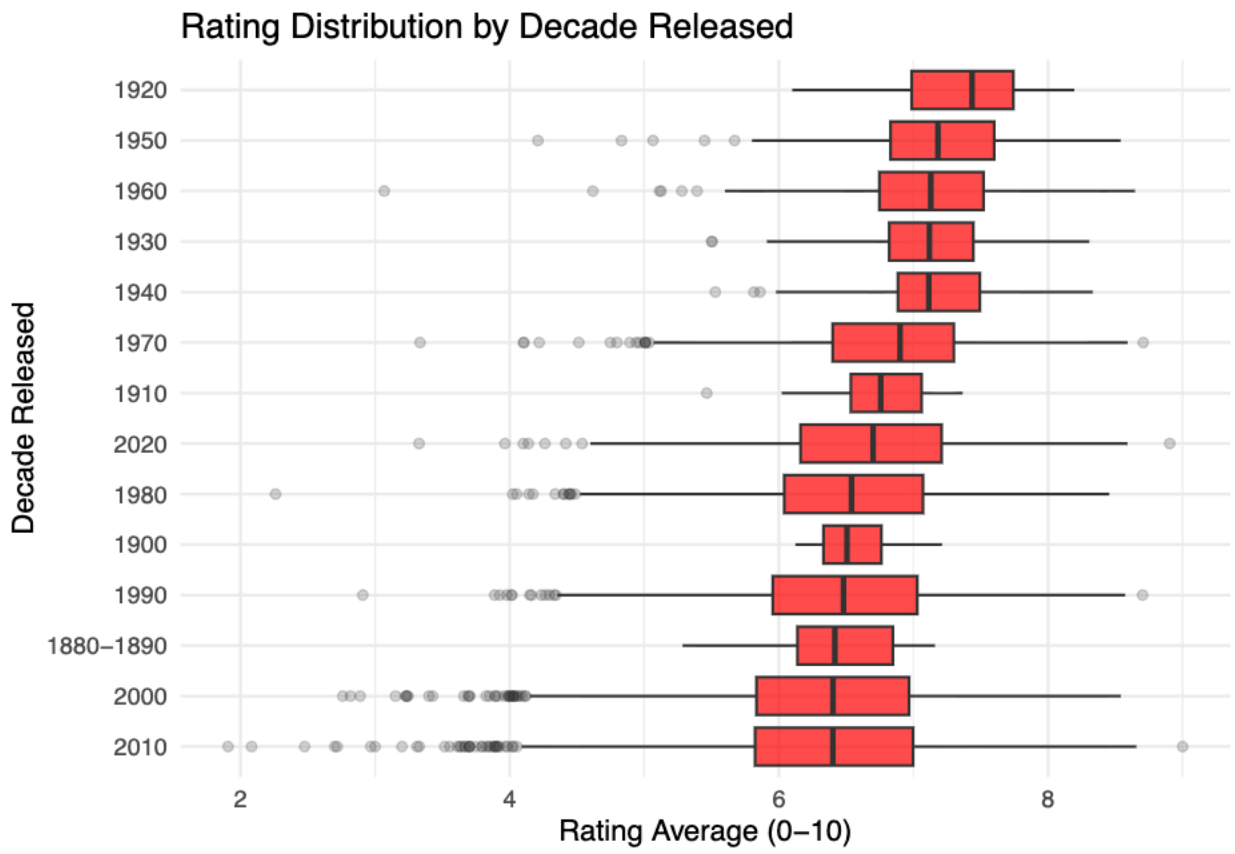
```r
movie_decades <- tmbd_movies_cleaned |>
  mutate(decade_released = (release_year %/% 10) * 10,
         decade_released = case_when(decade_released %in% c(1880, 1890) ~ "1880-1890",
         TRUE ~ as.character(decade_released)
         )
  )

movie_decades |>
  ggplot(aes(x = reorder(decade_released, roi_pct, FUN = median), y = roi_pct)) +
    geom_boxplot(outlier.alpha = 0.2, fill = "blue", alpha = 0.7) +
    coord_flip(ylim = c(-100, 1500)) +
    labs(
      title = "ROI Distribution by Decade Released",
```

```
    x = "Decade Released",
    y = "ROI %"
  ) +
  theme_minimal()
```




```
movie_decades |>
  ggplot(aes(x = reorder(decade_released, rating_average, FUN = median), y = rating_average)) +
   geom_boxplot(outlier.alpha = 0.2, fill = "red", alpha = 0.7) +
   coord_flip() +
   labs(
     title = "Rating Distribution by Decade Released",
     x = "Decade Released",
     y = "Rating Average (0-10)"
   ) +
   theme_minimal()
```



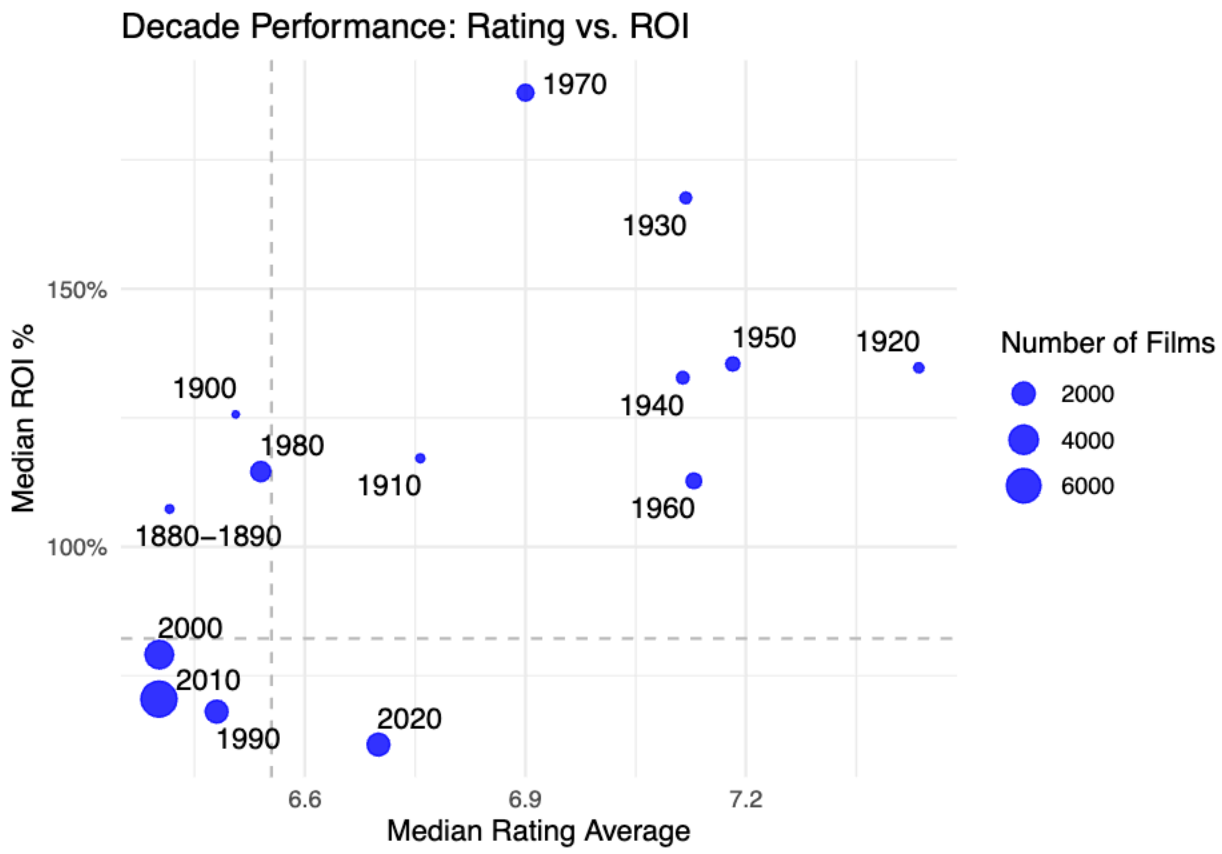

```
library(ggrepel)

decade_summary <- movie_decades |>
  group_by(decade_released) |>
  summarise(
    median_rating = median(rating_average, na.rm = TRUE),
    median_roi = median(roi_pct, na.rm = TRUE),
    movie_count = n()
  )


overall_median_rating <- median(tmbd_movies_cleaned$rating_average, na.rm = TRUE)
overall_median_roi <- median(tmbd_movies_cleaned$roi_pct, na.rm = TRUE)

ggplot(decade_summary, aes(x = median_rating, y = median_roi, label = decade_released)) +
  geom_vline(xintercept = overall_median_rating, linetype = "dashed", color = "gray") +
  geom_hline(yintercept = overall_median_roi, linetype = "dashed", color = "gray") +
  geom_point(aes(size = movie_count), color = "blue", alpha = 0.8) +
  geom_text_repel(size = 3.8, box.padding = 0.35) +
  scale_y_continuous(labels = scales::label_percent(scale = 1)) +
  labs(
    title = "Decade Performance: Rating vs. ROI",
    x = "Median Rating Average",
    y = "Median ROI %",
    size = "Number of Films"
  ) +
```

```
  theme_minimal()
```




```
decade_combined_performance <- movie_decades |>
  group_by(decade_released) |>
  summarise(
    movie_count = n(),
    median_rating = median(rating_average, na.rm = TRUE),
    median_roi = median(roi_pct, na.rm = TRUE)
  ) |>
  mutate(
    rating_percentile = percent_rank(median_rating),
    roi_percentile = percent_rank(median_roi),
    combined_genre_score = (rating_percentile + roi_percentile) / 2
  ) |>
  arrange(desc(combined_genre_score))

print(decade_combined_performance, n = Inf)
```

```
## # A tibble: 14 x 7
##    decade_released movie_count median_rating median_roi rating_percentile
##    <chr>                <int>         <dbl>      <dbl>             <dbl>
##  1 1920                    63          7.44       135.             1
##  2 1950                   364          7.18       135.             0.923
##  3 1930                   147          7.12       168.             0.769
##  4 1970                   736          6.9        188.             0.615
```

```
##  5 1940                  217          7.11      133.            0.692
##  6 1960                  550          7.13      113.            0.846
##  7 1910                   18          6.76      117.            0.538
##  8 1900                    4          6.51      126.            0.308
##  9 1980                 1309          6.54      115.            0.385
## 10 1880-1890              12          6.42      107.            0.154
## 11 2020                 1916          6.7        61.7           0.462
## 12 1990                 1937          6.48       68.0           0.231
## 13 2000                 3665          6.40       79.1           0.0769
## 14 2010                 6696          6.40       70.5           0
## # i 2 more variables: roi_percentile <dbl>, combined_genre_score <dbl>
```

For this data, due to there being only 3 entries in 1880s, I combined 1880 and 1890. From these results, we see the best movies have come from the 1920s and the highest ROI from the 1970s. The worst movies have come from the 2010s, but it is important to note that the most movies have come from 2010 which skews the data.

Does release season have an effect on ROI and rating?

```r
tmbd_movies_segmented <- tmbd_movies_cleaned |>
  mutate(
    release_window = case_when(
      is_holiday_release == 1 ~ "Holiday Release",
      is_summer_release == 1  ~ "Summer Release",
      TRUE                    ~ "Standard / Regular"
    )
  )

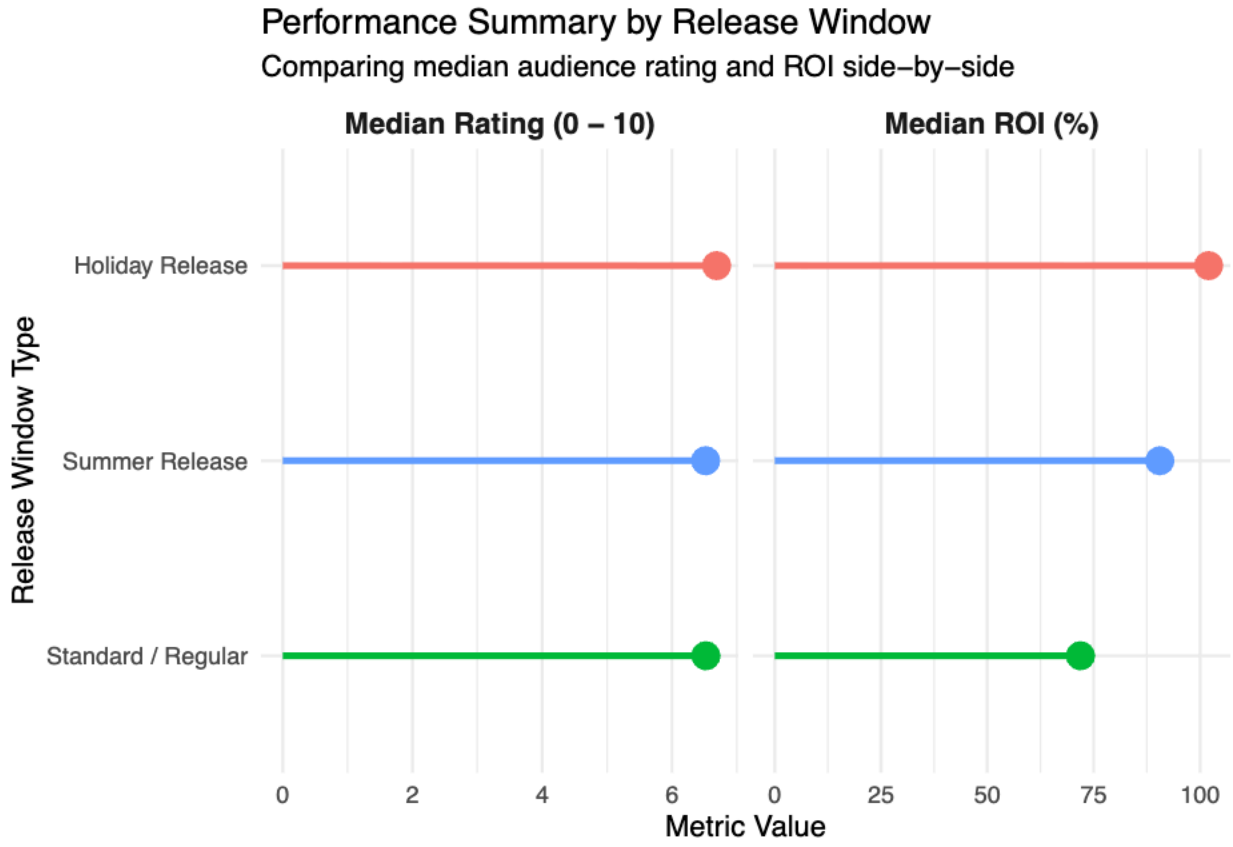
window_summary <- tmbd_movies_segmented |>
  filter(!is.na(release_window)) |>
  group_by(release_window) |>
  summarise(
    median_rating = median(rating_average, na.rm = TRUE),
    median_roi = median(roi_pct, na.rm = TRUE),
    movie_count = n()
  )

window_long <- window_summary |>
  pivot_longer(
    cols = c(median_rating, median_roi),
    names_to = "metric",
    values_to = "value"
  ) |>
  mutate(
    metric_label = if_else(metric == "median_rating", "Median Rating (0 - 10)", "Median ROI (%)")
  )

ggplot(window_long, aes(x = reorder(release_window, value), y = value, color = release_window)) +
  geom_segment(aes(xend = release_window, yend = 0), size = 1.2) +
  geom_point(size = 4.5) +
  coord_flip() +
  facet_wrap(~ metric_label, scales = "free_x") +
```

```r
  scale_y_continuous(labels = scales::label_comma()) +
  labs(
    title = "Performance Summary by Release Window",
    subtitle = "Comparing median audience rating and ROI side-by-side",
    x = "Release Window Type",
    y = "Metric Value"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    strip.text = element_text(face = "bold", size = 11)
  )
```

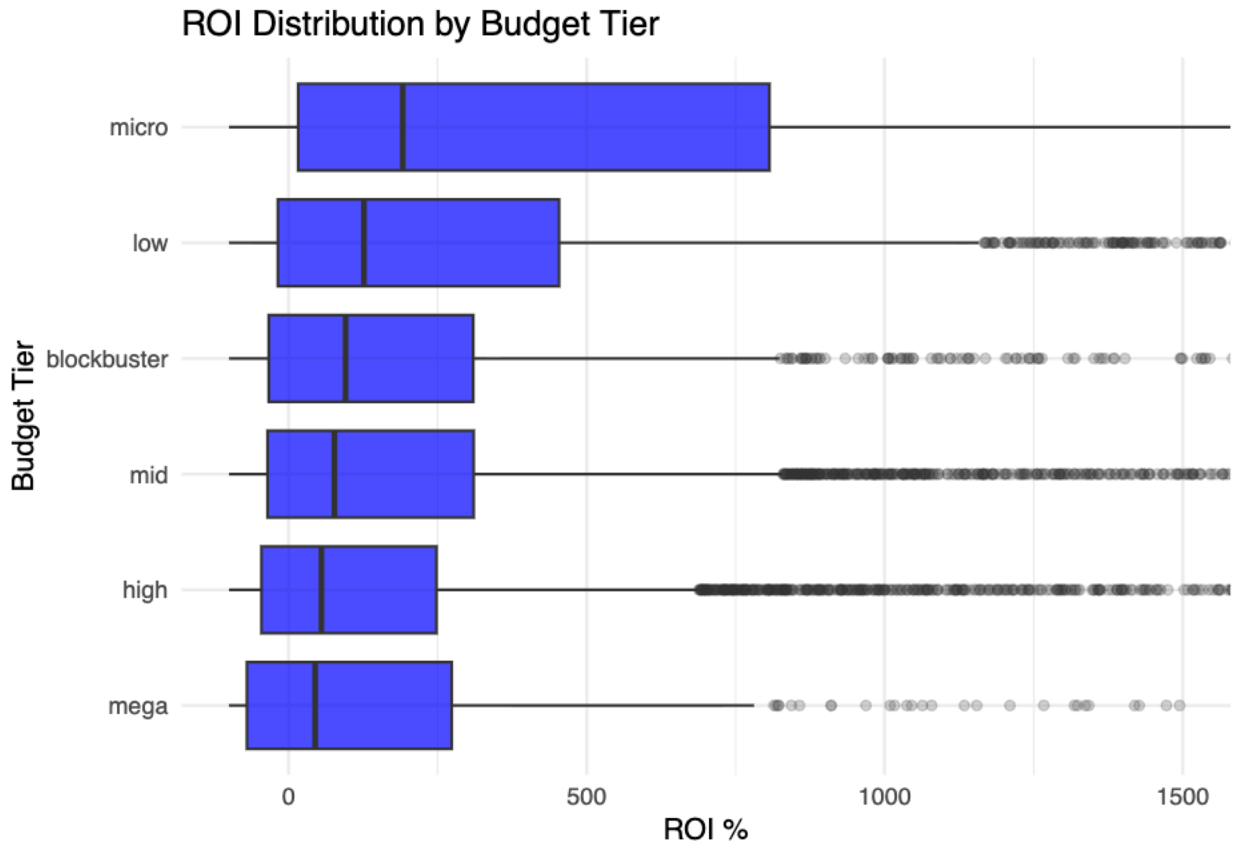
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



While the ratings are around the same, there is a stark contrast between holiday releases and the other releases in return on investment.

Does budget have an effect on rating and ROI?

```
tmbd_movies_cleaned |>
  ggplot(aes(x = reorder(budget_tier, roi_pct, FUN = median), y = roi_pct)) +
    geom_boxplot(outlier.alpha = 0.2, fill = "blue", alpha = 0.7) +
    coord_flip(ylim = c(-100, 1500)) +
    labs(
      title = "ROI Distribution by Budget Tier",
      x = "Budget Tier",
      y = "ROI %"
    ) +
    theme_minimal()
```




```
tmbd_movies_cleaned |>
   ggplot(aes(x = reorder(budget_tier, rating_average, FUN = median), y = rating_average)) +
    geom_boxplot(outlier.alpha = 0.2, fill = "red", alpha = 0.7) +
    coord_flip() +
    labs(
      title = "Rating Distribution by Budget Tier",
      x = "Budget Tier",
      y = "Rating Average (0-10)"
    ) +
    theme_minimal()
```




```
budget_combined_performance <- tmbd_movies_cleaned |>
  group_by(budget_tier) |>
  summarise(
    movie_count = n(),
    median_rating = median(rating_average, na.rm = TRUE),
    median_roi = median(roi_pct, na.rm = TRUE)
  ) |>
  filter(movie_count >= 50) |>
  mutate(
    rating_percentile = percent_rank(median_rating),
    roi_percentile = percent_rank(median_roi),
    combined_genre_score = (rating_percentile + roi_percentile) / 2
  ) |>
  arrange(desc(combined_genre_score))

print(budget_combined_performance, n = Inf)
```

```
## # A tibble: 6 x 7
##   budget_tier movie_count median_rating median_roi rating_percentile
##   <chr>             <int>         <dbl>      <dbl>             <dbl>
## 1 micro               920          6.82      192.                1
## 2 low                2926          6.69      126.                0.4
## 3 blockbuster        1876          6.7        95.5               0.6
## 4 mid                5638          6.5        76.8               0.2
## 5 mega                541          6.7        44.6               0.6
```

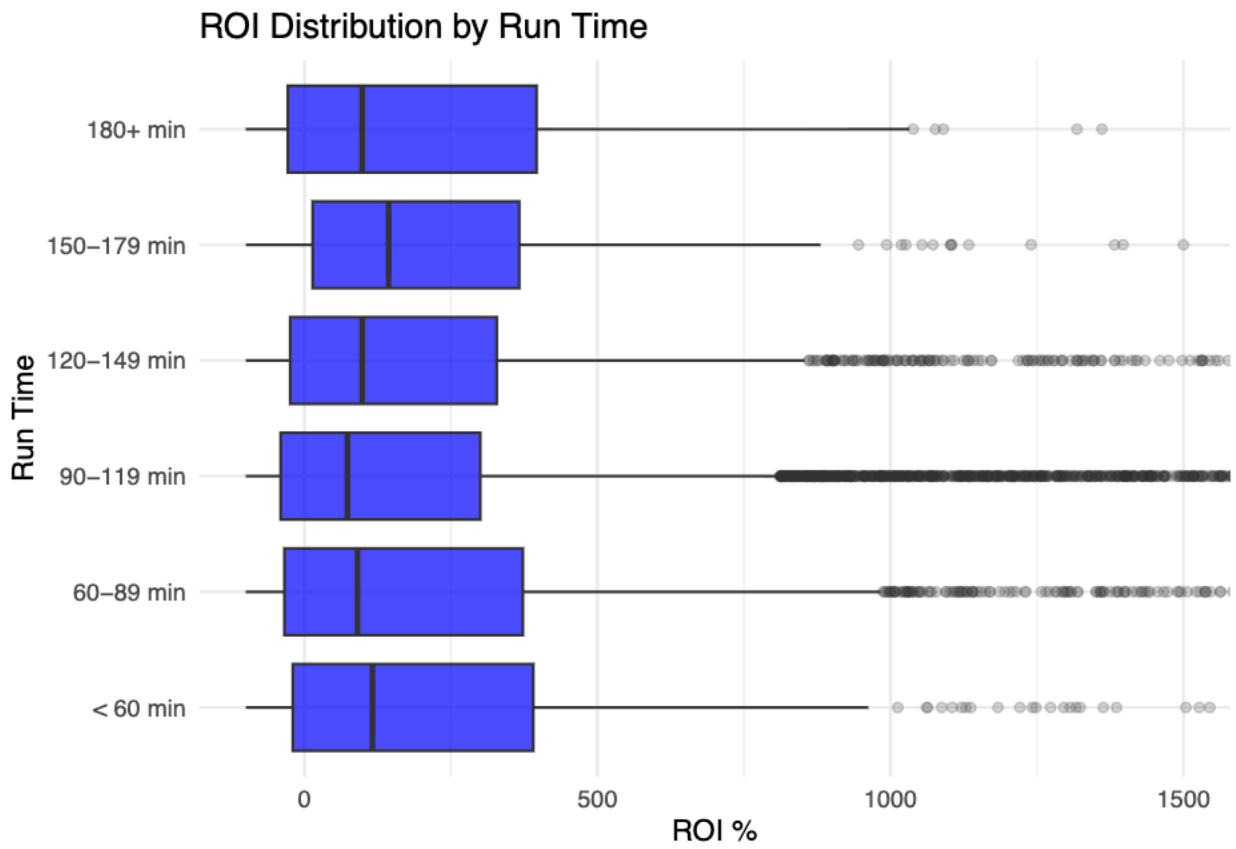
```
## 6 high               5733          6.46       55.0               0
## # i 2 more variables: roi_percentile <dbl>, combined_genre_score <dbl>
```

The results aren't the most surprising with the highest ROI being in micro budget tier as well as highest ratings in the micro tier, I believe this is due to the amount of movies with a vote count over 100 on these micro budget films had to be good to get popular. Due to this, the low budget tier I believe is statistically the best due to the amount of movies in that, and the combinded score.

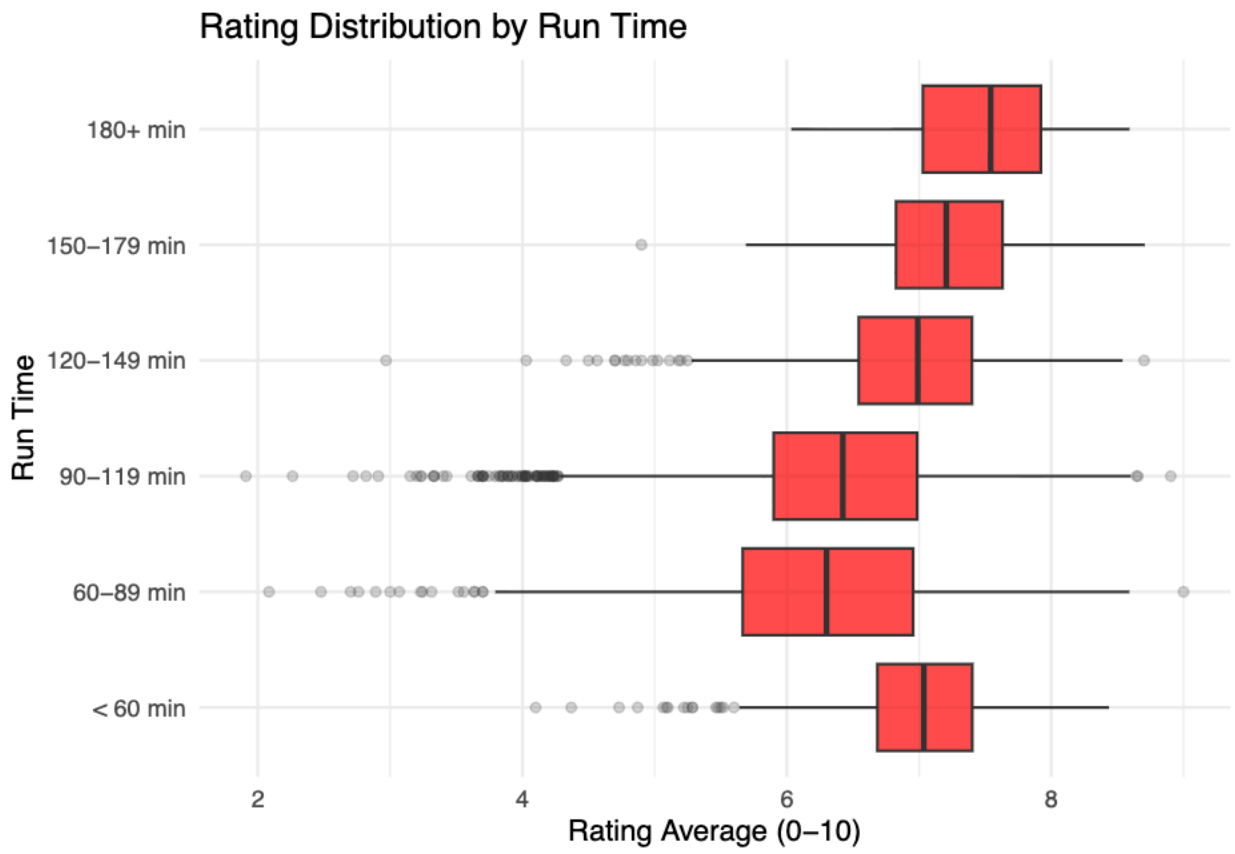
Does run-time have an effect on rating and ROI?

```
tmbd_movies_run_time <- tmbd_movies_cleaned |>
  mutate(
    run_time_segments = case_when(
      runtime_minutes >= 180 ~ "180+ min",
      runtime_minutes >= 150 ~ "150-179 min",
      runtime_minutes >= 120 ~ "120-149 min",
      runtime_minutes >= 90  ~ "90-119 min",
      runtime_minutes >= 60  ~ "60-89 min",
      TRUE                   ~ "< 60 min"
    ),
    # Force the exact left-to-right / bottom-to-top chronological display order
    run_time_segments = factor(
      run_time_segments,
      levels = c("< 60 min", "60-89 min", "90-119 min", "120-149 min", "150-179 min", "180+ min")
    )
  )

tmbd_movies_run_time |>
  ggplot(aes(x = run_time_segments, y = roi_pct)) +
    geom_boxplot(outlier.alpha = 0.2, fill = "blue", alpha = 0.7) +
    coord_flip(ylim = c(-100, 1500)) +
    labs(
      title = "ROI Distribution by Run Time",
      x = "Run Time",
      y = "ROI %"
    ) +
    theme_minimal()
```

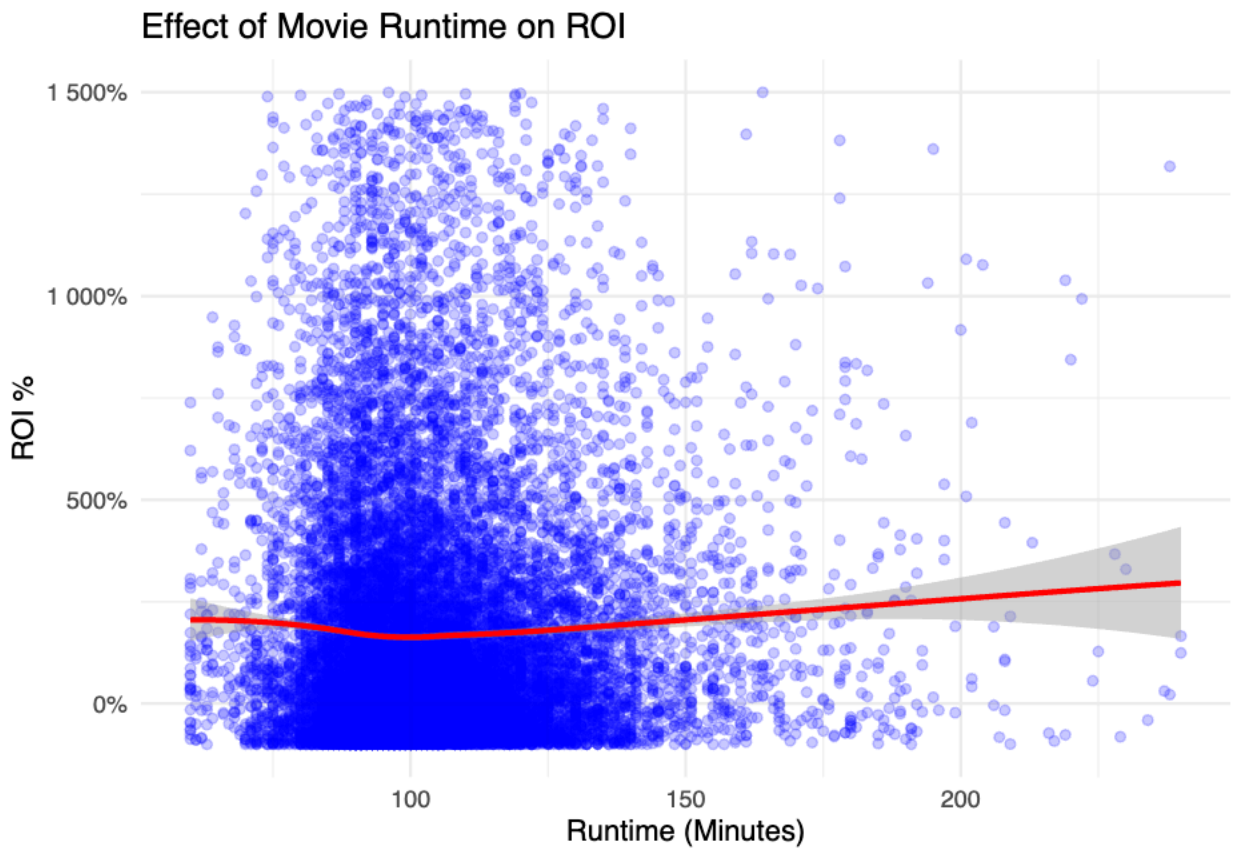



```
tmbd_movies_run_time |>
   ggplot(aes(x = run_time_segments, y = rating_average)) +
    geom_boxplot(outlier.alpha = 0.2, fill = "red", alpha = 0.7) +
    coord_flip() +
    labs(
      title = "Rating Distribution by Run Time",
      x = "Run Time",
      y = "Rating Average (0-10)"
    ) +
    theme_minimal()
```



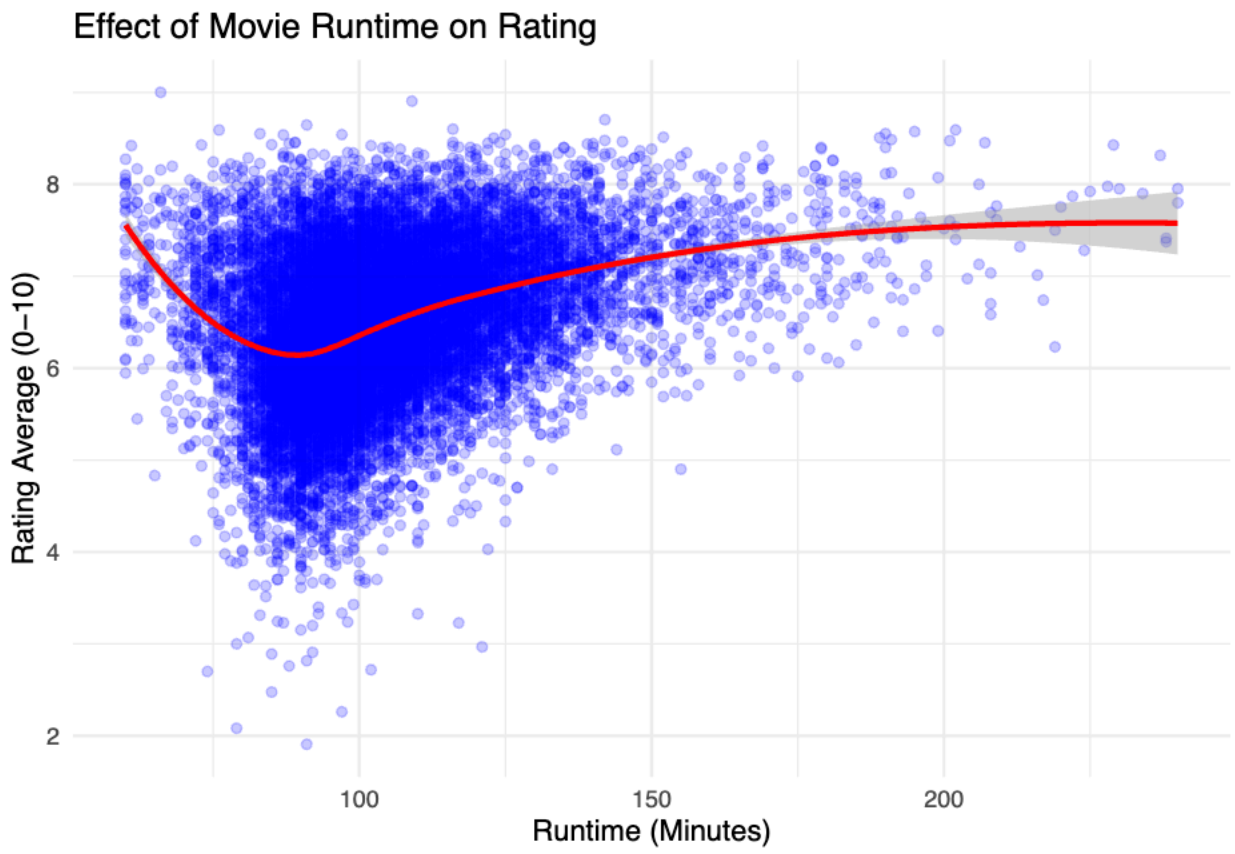

```
tmbd_movies_cleaned |>
  filter(runtime_minutes >= 60, runtime_minutes <= 240, roi_pct <= 1500) |>
  ggplot(aes(x = runtime_minutes, y = roi_pct)) +
  geom_point(alpha = 0.2, color = "blue") +
  geom_smooth(method = "loess", color = "red", se = TRUE) +
  scale_y_continuous(labels = scales::label_percent(scale = 1)) +
  labs(
    title = "Effect of Movie Runtime on ROI",
    x = "Runtime (Minutes)",
    y = "ROI %"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
tmbd_movies_cleaned |>
  filter(runtime_minutes >= 60, runtime_minutes <= 240, roi_pct <= 1500) |>
  ggplot(aes(x = runtime_minutes, y = rating_average)) +
  geom_point(alpha = 0.2, color = "blue") +
  geom_smooth(method = "loess", color = "red", se = TRUE) +
  labs(
    title = "Effect of Movie Runtime on Rating",
    x = "Runtime (Minutes)",
    y = "Rating Average (0-10)"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
run_time_combined_performance <- tmbd_movies_run_time |>
  group_by(run_time_segments) |>
  summarise(
    movie_count = n(),
    median_rating = median(rating_average, na.rm = TRUE),
    median_roi = median(roi_pct, na.rm = TRUE)
  ) |>
  filter(movie_count >= 50) |>
  mutate(
    rating_percentile = percent_rank(median_rating),
    roi_percentile = percent_rank(median_roi),
    combined_genre_score = (rating_percentile + roi_percentile) / 2
  ) |>
  arrange(desc(combined_genre_score))

print(run_time_combined_performance, n = Inf)
```

```
## # A tibble: 6 x 7
##   run_time_segments movie_count median_rating median_roi rating_percentile
##   <fct>                   <int>         <dbl>      <dbl>             <dbl>
## 1 150-179 min               365          7.21      144.               0.8
## 2 180+ min                  132          7.54       98.5              1
## 3 < 60 min                  522          7.04      116.               0.6
## 4 120-149 min              2494          6.99       98.4              0.4
## 5 60-89 min                3010          6.3        90.4              0
```

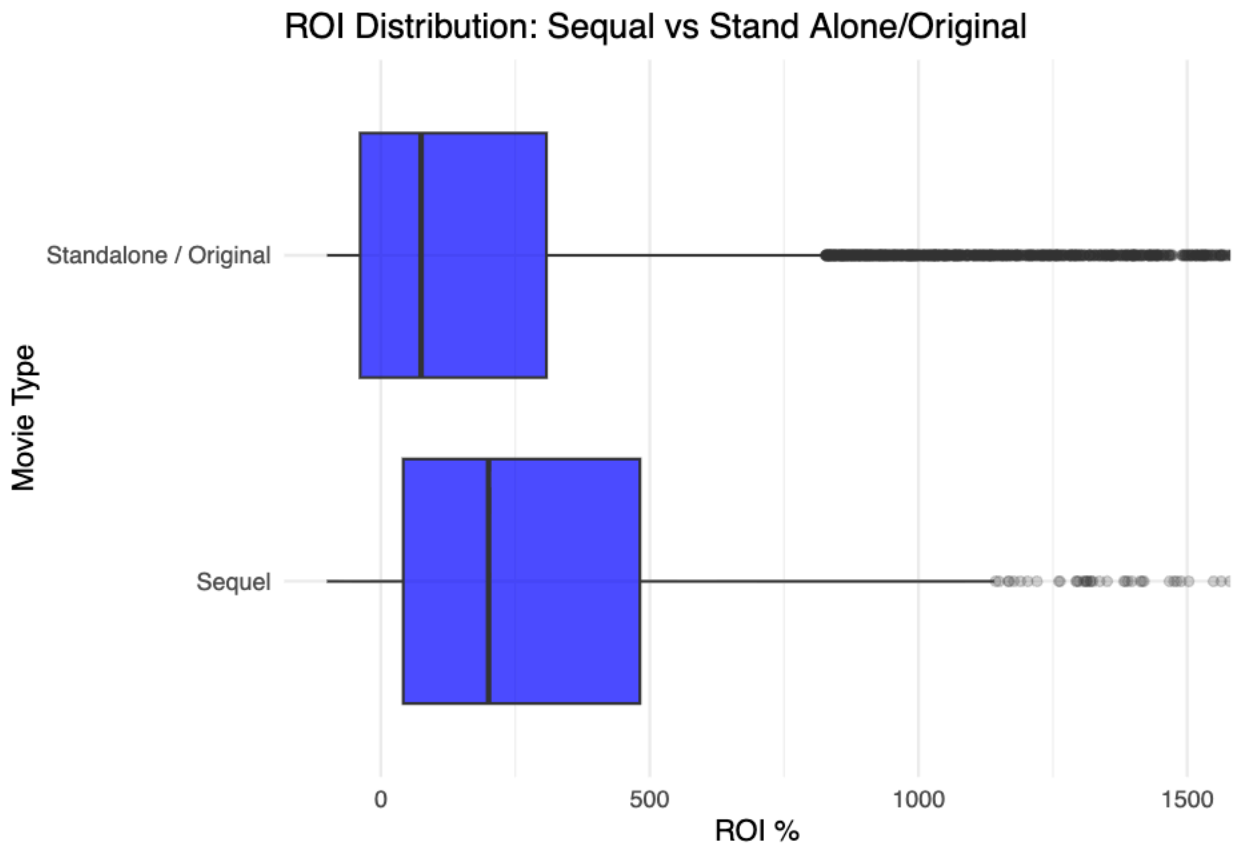
```
## 6 90-119 min              11111          6.42       73.3               0.2
## # i 2 more variables: roi_percentile <dbl>, combined_genre_score <dbl>
```

Yes, there is an effect of run time on rating and ROI. From the box plots as well as the scatter plots, as the movie gets longer, the average rating increases as well as the ROI, up until 180 minutes for the ROI. Based on this information, a movie with run time between 150-179 minutes has the best combined results.

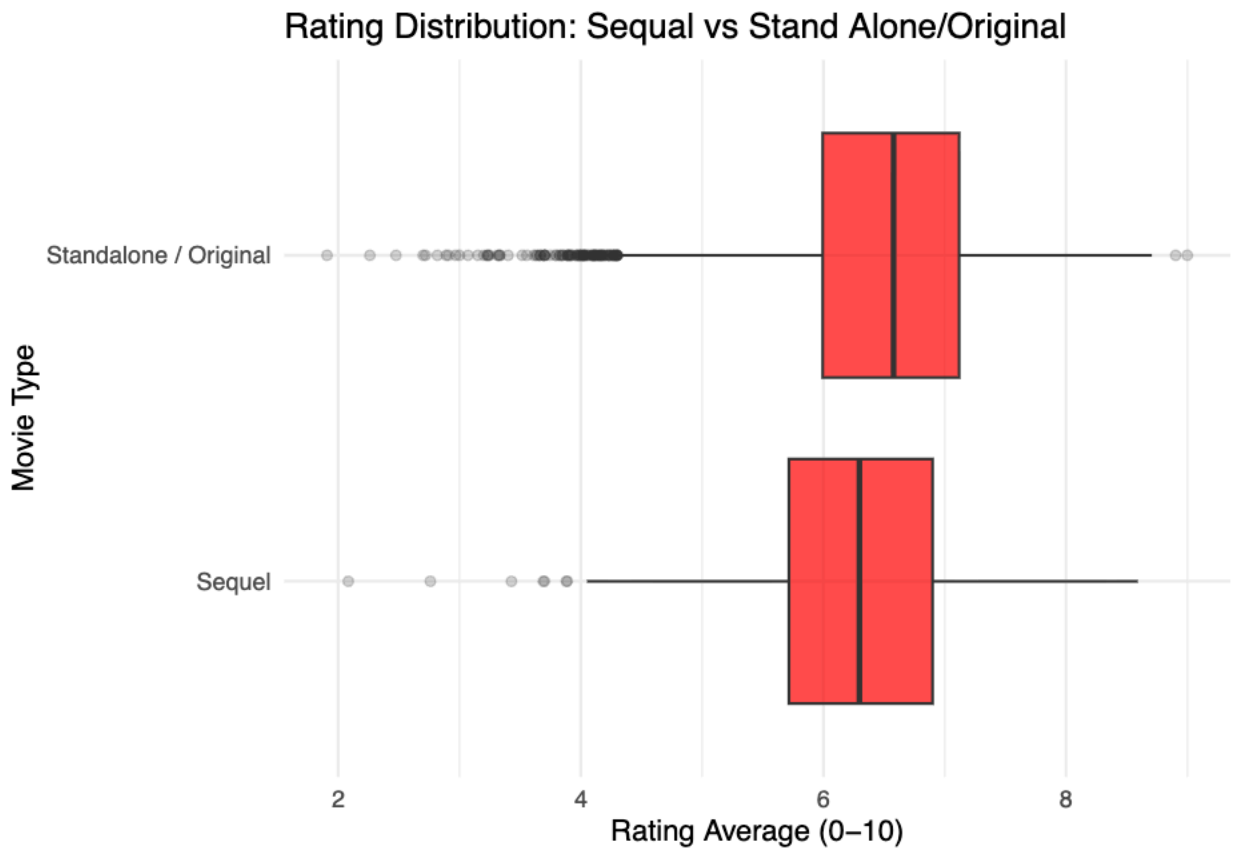
Are sequels on average "good" movies (based on rating and ROI)?

```
tmbd_movies_sequels <- tmbd_movies_cleaned |>
  mutate(
    movie_type = case_when(
      is_sequel == 1 ~ "Sequel",
      TRUE ~ "Standalone / Original"
    )
  )

tmbd_movies_sequels |>
  ggplot(aes(x = movie_type, y = roi_pct)) +
    geom_boxplot(outlier.alpha = 0.2, fill = "blue", alpha = 0.7) +
    coord_flip(ylim = c(-100, 1500)) +
    labs(
      title = "ROI Distribution: Sequal vs Stand Alone/Original",
      x = "Movie Type",
      y = "ROI %"
    ) +
    theme_minimal()
```




```
tmbd_movies_sequels |>
   ggplot(aes(x = movie_type, y = rating_average)) +
    geom_boxplot(outlier.alpha = 0.2, fill = "red", alpha = 0.7) +
    coord_flip() +
    labs(
      title = "Rating Distribution: Sequal vs Stand Alone/Original",
      x = "Movie Type",
      y = "Rating Average (0-10)"
    ) +
    theme_minimal()
```



Based on these results, sequels are profitable movies, but have a worse rating than stand alone/originals on average.

What are the Top 10 Movies of all time based on ROI and average rating combined?

```
top_10_balanced_movies <- tmbd_movies_cleaned |>
  mutate(
    roi_percentile = percent_rank(roi_pct),
    rating_percentile = percent_rank(rating_average),
    combined_score = (roi_percentile + rating_percentile) / 2
  ) |>
  arrange(desc(combined_score)) |>
  slice_head(n = 20) |>
  select(title, release_year, primary_genre, rating_average, roi_pct, combined_score)

print(top_10_balanced_movies)
```

```
##                                          title release_year
## 1                                The Godfather         1972
## 2                                       Psycho         1960
## 3                                    Star Wars         1977
## 4                One Flew Over the Cuckoo's Nest         1975
## 5                 The Good, the Bad and the Ugly         1966
## 6                                   Rear Window         1954
## 7                Me Against You: Mr. S's Vendetta         2020
```

```
## 8                                          Your Name.         2016
## 9                                The Empire Strikes Back         1980
## 10                                  In the Mood for Love         2000
## 11                               Ayla: The Daughter of War         2017
## 12                                    The Intouchables         2011
## 13 Demon Slayer -Kimetsu no Yaiba- The Movie: Mugen Train         2020
## 14                                          Better Days         2019
## 15                                         Pulp Fiction         1994
## 16                 Rascal Does Not Dream of a Dreaming Girl         2019
## 17                                      Sansho the Bailiff         1954
## 18                               Michael Jackson's Thriller         1983
## 19                                            Parasite         2019
## 20                                      Gone with the Wind         1939
##    primary_genre rating_average  roi_pct combined_score
## 1          Drama          8.707  3984.44      0.9954914
## 2         Horror          8.437  3865.56      0.9943005
## 3      Adventure          8.204  6949.07      0.9938468
## 4          Drama          8.420  3532.71      0.9932513
## 5        Western          8.473  3141.67      0.9928827
## 6       Thriller          8.359  3603.45      0.9927976
## 7         Family          8.130 12566.67      0.9925707
## 8        Romance          8.514  2853.89      0.9918335
## 9      Adventure          8.400  2891.11      0.9910112
## 10         Drama          8.100  9369.75      0.9909828
## 11         Drama          8.090  9358.53      0.9906425
## 12         Drama          8.277  3181.45      0.9905858
## 13        Action          8.259  3083.95      0.9899336
## 14         Drama          8.183  3554.38      0.9893665
## 15      Thriller          8.488  2416.47      0.9890546
## 16     Animation          8.200  2987.75      0.9882606
## 17         Drama          8.138  3324.79      0.9873816
## 18        Horror          8.163  3020.50      0.9872965
## 19        Comedy          8.515  2166.93      0.9871548
## 20         Drama          7.965  9958.81      0.9860489
```

What are the Top 10 Sequels of all time based on ROI and average rating combined?

```
top_10_balanced_movies_sequel <- tmbd_movies_cleaned |>
  filter(is_sequel == 1) |>
  mutate(
    roi_percentile = percent_rank(roi_pct),
    rating_percentile = percent_rank(rating_average),
    combined_score = (roi_percentile + rating_percentile) / 2
  ) |>
  arrange(desc(combined_score)) |>
  slice_head(n = 10) |>
  select(title, release_year, primary_genre, rating_average, roi_pct, combined_score)

print(top_10_balanced_movies_sequel)
```

```
##                                    title release_year primary_genre
## 1 The Empire Strikes Back         1980     Adventure
```

```
## 2                         Miracle in Cell No. 7         2013         Drama
## 3                       My Mom Is a Character 3         2019        Comedy
## 4      The Human Condition II: Road to Eternity         1959           War
## 5                       My Mom Is a Character 2         2016        Comedy
## 6                                   Interstate 60         2002     Adventure
## 7  Harry Potter and the Deathly Hallows: Part 2         2011       Fantasy
## 8                                The Mountain II         2016           War
## 9                                     Stalag 17         1953        Comedy
## 10                                        Split         2017        Horror
##    rating_average roi_pct combined_score
## 1           8.400 2891.11      0.9876135
## 2           7.926 2671.61      0.9744013
## 3           8.166 1863.65      0.9698596
## 4           8.200 1382.16      0.9603633
## 5           7.655 1792.91      0.9475640
## 6           7.400 8016.83      0.9417836
## 7           8.105  973.21      0.9372419
## 8           7.506 1598.34      0.9339389
## 9           7.707 1149.68      0.9322874
## 10          7.337 2993.94      0.9277457
```

Prediction Model

In conclusion, the best movie statistically, based on average rating and return on investment, would be a movie made in 1920, produced in Mexico, the genre would be a documentary, it would be released during the holiday season, with a low budget tier, and a run time of 150-179 minutes. For an added bonus to this conclusion, I will make a prediction model on the projected ROI and rating of a movie of this nature.

```
model_data <- tmbd_movies_cleaned |>
  mutate(
    release_window = case_when(
      is_holiday_release == 1 ~ "Holiday Release",
      is_summer_release == 1  ~ "Summer Release",
      TRUE                    ~ "Standard / Regular"
    ),
    roi_percentile = percent_rank(roi_pct),
    rating_percentile = percent_rank(rating_average),
    combined_score = (roi_percentile + rating_percentile) / 2,
    primary_genre = as.factor(primary_genre),
    primary_production_country = as.factor(primary_production_country),
    budget_tier = as.factor(budget_tier),
    release_window = as.factor(release_window)
  ) |>
  filter(
    !is.na(combined_score),
    !is.na(budget_tier),
    !is.na(primary_production_country)
  )

rating_model <- lm(
  rating_average ~ primary_genre + primary_production_country + budget_tier + runtime_minutes + release
  data = model_data
```

```
)


roi_model <- lm(
  roi_pct ~ primary_genre + primary_production_country + budget_tier + runtime_minutes + release_window
  data = model_data
)


combined_model <- lm(
  combined_score ~ primary_genre + primary_production_country + budget_tier + runtime_minutes + release_
  data = model_data
)

hypothetical_movie <- data.frame(
  release_year = 1920,
  primary_genre = factor("Documentary", levels = levels(model_data$primary_genre)),
  primary_production_country = factor("Mexico", levels = levels(model_data$primary_production_country))
  budget_tier = factor("low", levels = levels(model_data$budget_tier)),
  runtime_minutes = 165,
  release_window = factor("Holiday Release", levels = levels(model_data$release_window))
)

predicted_rating <- predict(rating_model, newdata = hypothetical_movie)
predicted_roi <- predict(roi_model, newdata = hypothetical_movie)
predicted_score <- predict(combined_model, newdata = hypothetical_movie)


print(predicted_score)
```

```
##        1
## 1.077525
```

```
print(predicted_roi)
```

```
##         1
## -49.74769
```

```
print(predicted_rating)
```

```
##        1
## 9.154542
```

This model produced very interesting results, outputting a combined score prediction of over 1 which is mathematically impossible, it produced an amazing rating of 9.1 which I expected, but it produced a negative 50% return on investment, which makes sense because we are using an outlier year in 1920 with very few data points that the prediction model can base itself off of. I am going to run this model with 1950 due to the larger sample size and because it is ranked as number 2 in combined score for decades.

```r
model_data <- tmbd_movies_cleaned |>
  mutate(
    release_window = case_when(
      is_holiday_release == 1 ~ "Holiday Release",
      is_summer_release == 1  ~ "Summer Release",
      TRUE                    ~ "Standard / Regular"
    ),
    roi_percentile = percent_rank(roi_pct),
    rating_percentile = percent_rank(rating_average),
    combined_score = (roi_percentile + rating_percentile) / 2,
    primary_genre = as.factor(primary_genre),
    primary_production_country = as.factor(primary_production_country),
    budget_tier = as.factor(budget_tier),
    release_window = as.factor(release_window)
  ) |>
  filter(
    !is.na(combined_score),
    !is.na(budget_tier),
    !is.na(primary_production_country)
  )

rating_model <- lm(
  rating_average ~ primary_genre + primary_production_country + budget_tier + runtime_minutes + release
  data = model_data
)


roi_model <- lm(
  roi_pct ~ primary_genre + primary_production_country + budget_tier + runtime_minutes + release_window
  data = model_data
)


combined_model <- lm(
  combined_score ~ primary_genre + primary_production_country + budget_tier + runtime_minutes + release
  data = model_data
)

hypothetical_movie <- data.frame(
  release_year = 1950,
  primary_genre = factor("Documentary", levels = levels(model_data$primary_genre)),
  primary_production_country = factor("Mexico", levels = levels(model_data$primary_production_country))
  budget_tier = factor("low", levels = levels(model_data$budget_tier)),
  runtime_minutes = 165,
  release_window = factor("Holiday Release", levels = levels(model_data$release_window))
)

predicted_rating <- predict(rating_model, newdata = hypothetical_movie)
predicted_roi <- predict(roi_model, newdata = hypothetical_movie)
predicted_score <- predict(combined_model, newdata = hypothetical_movie)


print(predicted_score)
```

```
##        1
## 1.012026
```

```
print(predicted_roi)
```

```
##         1
## -70.29145
```

```
print(predicted_rating)
```

```
##        1
## 8.877109
```

Shockingly this outputted a worse return on investment, I am now going to attempt 1970 as it has an even larger data set to base the model off of.

```
model_data <- tmbd_movies_cleaned |>
  mutate(
    release_window = case_when(
      is_holiday_release == 1 ~ "Holiday Release",
      is_summer_release == 1  ~ "Summer Release",
      TRUE                    ~ "Standard / Regular"
    ),
    roi_percentile = percent_rank(roi_pct),
    rating_percentile = percent_rank(rating_average),
    combined_score = (roi_percentile + rating_percentile) / 2,
    primary_genre = as.factor(primary_genre),
    primary_production_country = as.factor(primary_production_country),
    budget_tier = as.factor(budget_tier),
    release_window = as.factor(release_window)
  ) |>
  filter(
    !is.na(combined_score),
    !is.na(budget_tier),
    !is.na(primary_production_country)
  )

rating_model <- lm(
  rating_average ~ primary_genre + primary_production_country + budget_tier + runtime_minutes + release_
  data = model_data
)


roi_model <- lm(
  roi_pct ~ primary_genre + primary_production_country + budget_tier + runtime_minutes + release_window
  data = model_data
)


combined_model <- lm(
  combined_score ~ primary_genre + primary_production_country + budget_tier + runtime_minutes + release_
  data = model_data
```

```
)

hypothetical_movie <- data.frame(
  release_year = 1970,
  primary_genre = factor("Documentary", levels = levels(model_data$primary_genre)),
  primary_production_country = factor("Mexico", levels = levels(model_data$primary_production_country)),
  budget_tier = factor("low", levels = levels(model_data$budget_tier)),
  runtime_minutes = 165,
  release_window = factor("Holiday Release", levels = levels(model_data$release_window))
)

predicted_rating <- predict(rating_model, newdata = hypothetical_movie)
predicted_roi <- predict(roi_model, newdata = hypothetical_movie)
predicted_score <- predict(combined_model, newdata = hypothetical_movie)


print(predicted_score)
```

```
##         1
## 0.9683609
```

```
print(predicted_roi)
```

```
##         1
## -83.98728
```

```
print(predicted_rating)
```

```
##        1
## 8.692153
```

These results are shocking, but it shows the failures that can come with using a prediction model based on historical data. Due to there being no exact film to base the prediction model off of as well it causes these distortions in the data.

Conclusion

Thank you for joining me in this exploratory data analysis journey! I hope you learned something new!